

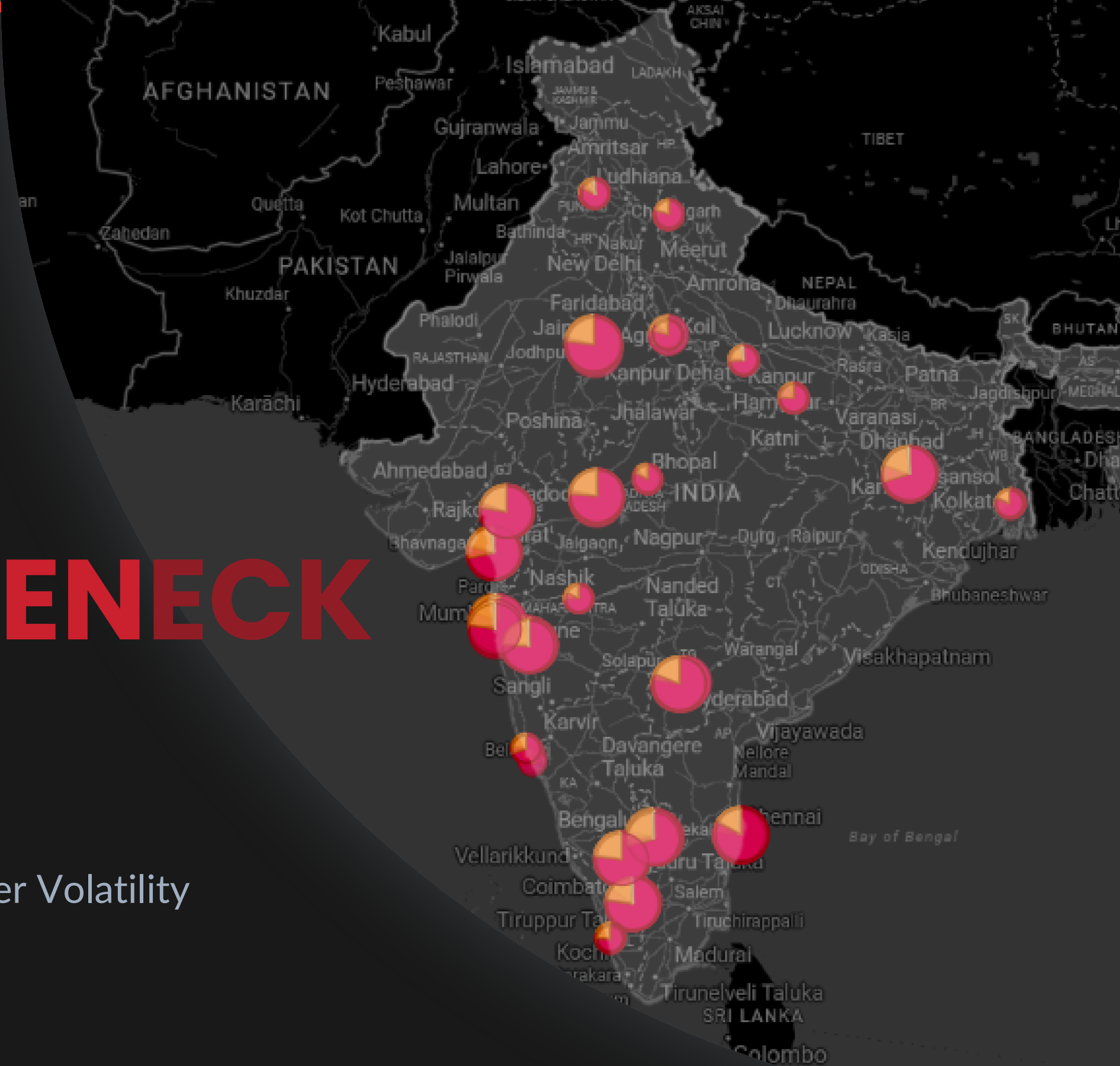
ZOMATO INDIA: LOGISTICS BOTTLENECK ANALYSIS

Uncovering the Impact of Delivery Delays on Daily Order Volatility

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By: Muhammad Rafly Alviansyah





Rafly's Website Portfolio



About me

1. Best Graduate in Geography Education at Universitas Pendidikan Indonesia, currently pivoting my career path into Data Analytics.

2. Incoming Retail Operations Data Analyst Intern at Hypefast, applying data-driven strategies to optimize retail performance and operational workflows.

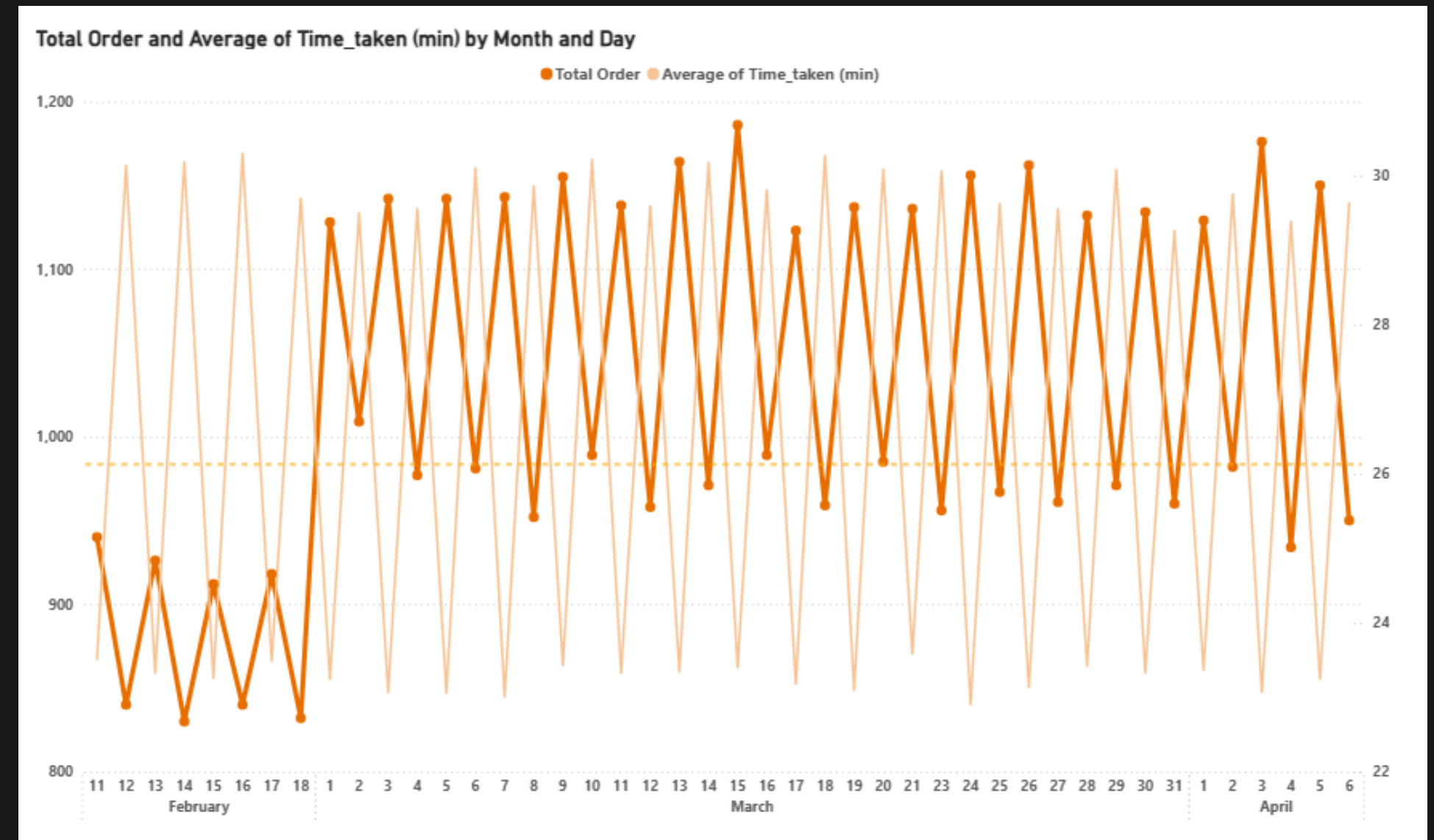
3. Highly passionate about Operations & Logistics Analytics, focusing on optimizing delivery performance, resolving supply-demand bottlenecks, and enhancing customer satisfaction.



BUSINESS CONTEXT

The Analytical Challenge

Management needs to diagnose the root cause of daily order stagnation. Initial operational trend shows a strong inverse pattern: spikes in delivery duration (Time Taken) directly suppress daily order volume.



Key Business Questions

1. What specific factors (e.g., traffic density, multi-batching) cause daily delivery duration to fluctuate so aggressively?
2. At what delivery duration threshold does customer rating begin to collapse, driving order volume down?
3. How should Zomato rebalance driver allocation across city categories to stabilize delivery time and unlock order growth?

METHODOLOGY & DATA PROCESSING



Data Scope & Cleaning

- Analyzed **45k** transaction records and **1.32K** active delivery partners.
- Standardized city typologies (e.g., corrected 'Metropolitan' to 'Metropolitan') to ensure total data integrity.



Feature Engineering

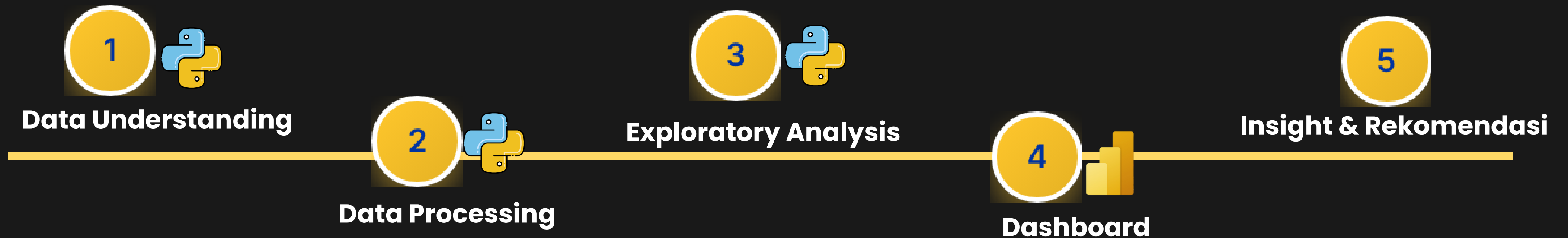
- Calculated exact delivery distances (distance_km) using the Haversine Formula from coordinates.
- Grouped 'Multiple Deliveries' into distinct batch tiers (Single, 1 Batch, 2 Batch, 3 Batch) and encoded traffic/weather variables into numeric indices.



Daily Aggregation

Applied Daily Aggregation specifically for March (the most complete dataset) to eliminate row-level noise and accurately uncover macro-business correlation trends.

Flow **Analysis Data**



Dataset Overview

- Model Architecture: Single Flat Transactional Schema (processed into a Virtual Star-Schema in Power BI for file optimization).
- Granularity: Level Transaksi (Row-level per single delivery order).
- Primary Key: Unique Transaction ID.
- Core Derived Metrics:
 - **distance_km** : Calculated via Haversine Formula using Restaurant and Delivery coordinates.
 - **multiple_deliveries** : Grouped into distinct Batch Order tiers (Single, 1 Batch, 2 Batch, 3 Batch)
 - **Order_Day_Name & Daily_Aggregated** : Feature engineered via DAX & Python Pandas for macro-level trend analysis..

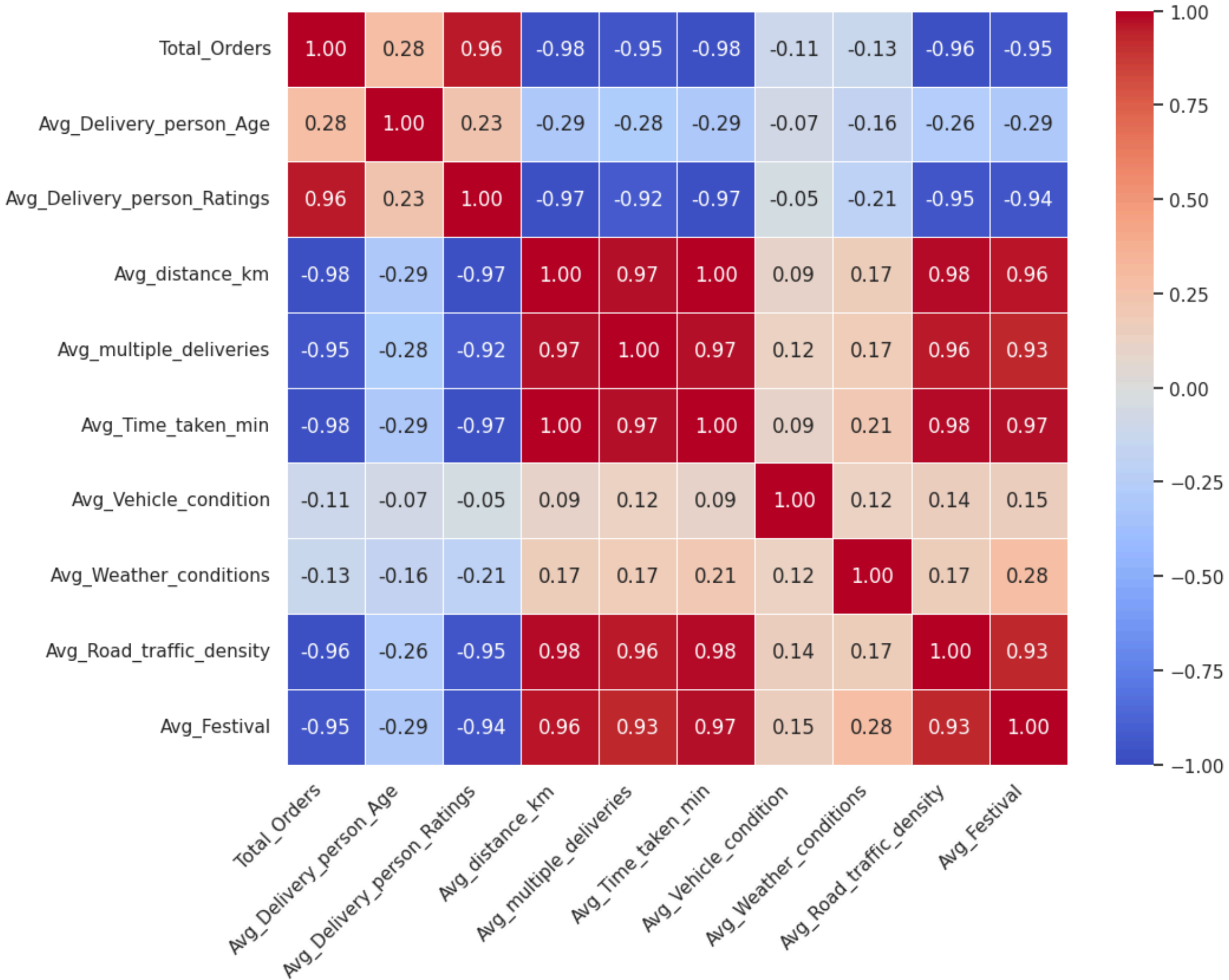
Table Name (Virtual Schema)	Primary Key & Key Features
zomato_deliveries (Main Dataset)	ID (Primary Key), Delivery_person_ID, Order_Date, Time_Orderd
spatial_dimension (Inferred Geo)	Restaurant_latitude/longitude, Delivery_location_latitude/longitude, City (Metropolitan, Urban, Semi-Urban)
driver_dimension (Inferred Profile)	Delivery_person_Age, Delivery_person_Ratings
logistics_metrics (Engineered Features)	Time_taken (min), distance_km, multiple_deliveries, Road_traffic_density

45K TOTAL Rows

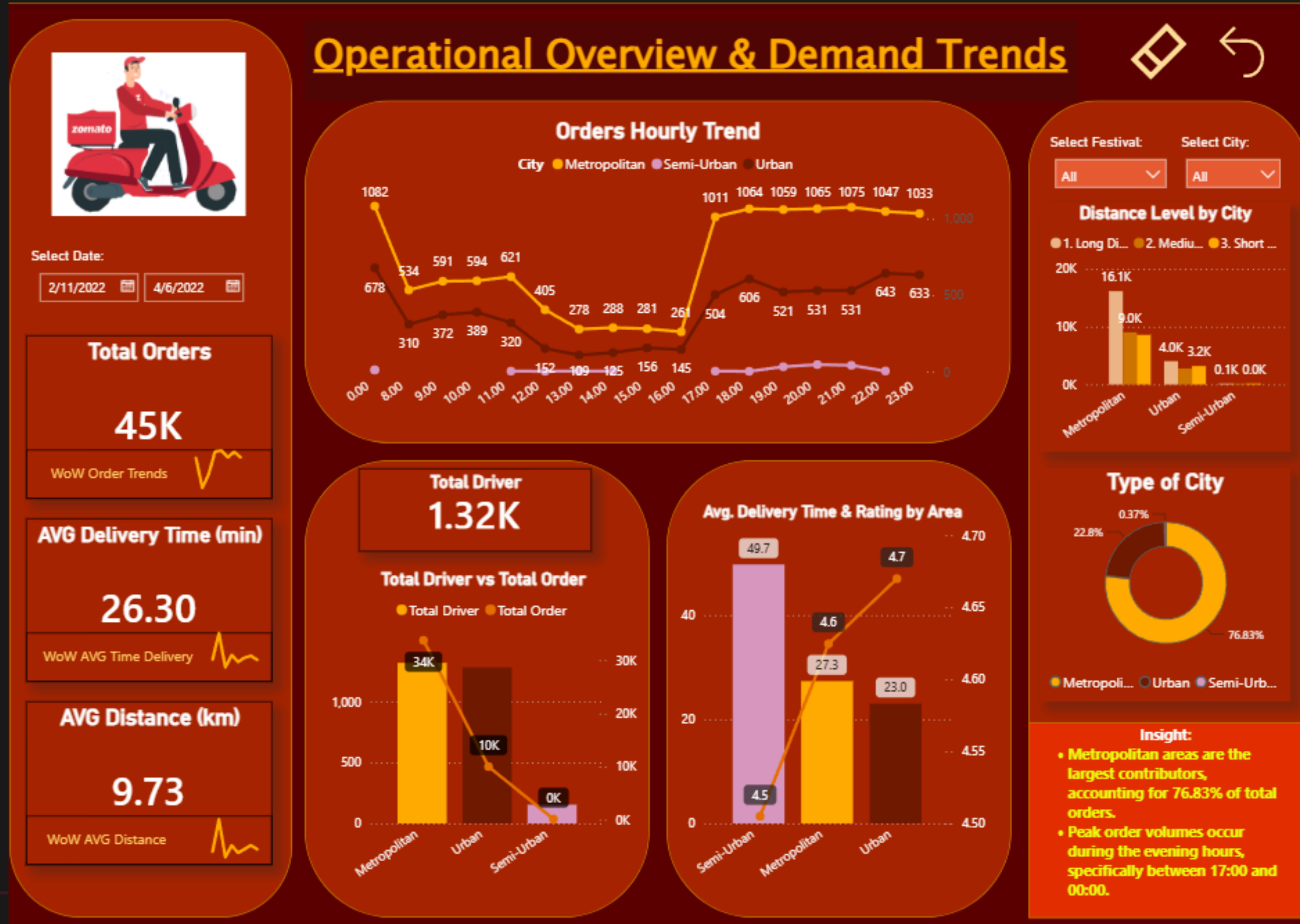
01. EDA: Operational Overview

Understanding the Macro Scale of Zomato's Network

Daily Matriks Korelasi Zomato: Khusus Bulan Maret



Preview Dashboard(1/3)



45K

TOTAL ORDERS PROCESSED

Successfully handled across the entire logistics network.

76%

METROPOLITAN DOMINANCE

Accounts for 34K total orders, making it the absolute core market.

17:00

PEAK DEMAND SURGE

Massive spikes consistently occur between 5 PM and Midnight.

THE DISCOVERY: ORDERS VS. RATINGS

The Synchronization

By analyzing the daily trend line across March, we discovered a perfect synchronization. The daily count of orders moves exactly in tandem with the Average Delivery Ratings.

This indicates that customer satisfaction has an immediate, same-day impact on retention and re-order rates.

Statistical Proof

We executed a Daily Aggregation Correlation Matrix (specifically for March to remove missing data bias). The results were definitive:

Correlation between Total Orders and Avg Rating is
+0.96

THE CORE BOTTLENECK

If Ratings drive Orders, what destroys Ratings?

The correlation matrix revealed massive negative impacts on daily volume. Stagnation is absolutely caused by operational delays.

- **Delayed Deliveries:** Correlation vs Total Orders is **-0.98**
- **Heavy Traffic Density:** Correlation vs Total Orders is **-0.96**

Conclusion: Drivers get stuck, delivery takes longer, ratings plummet, and order volume stagnates.



02. Diagnosing The Delays

Breaking down the operational failures step-by-step

Preview Dashboard(2/3)



Select Date:

2/11/2022

4/6/2022

Total Orders

45K

WoW Order Trends



AVG Delivery Time (min)

26.30

WoW AVG Time Delivery



AVG Rating

4.64

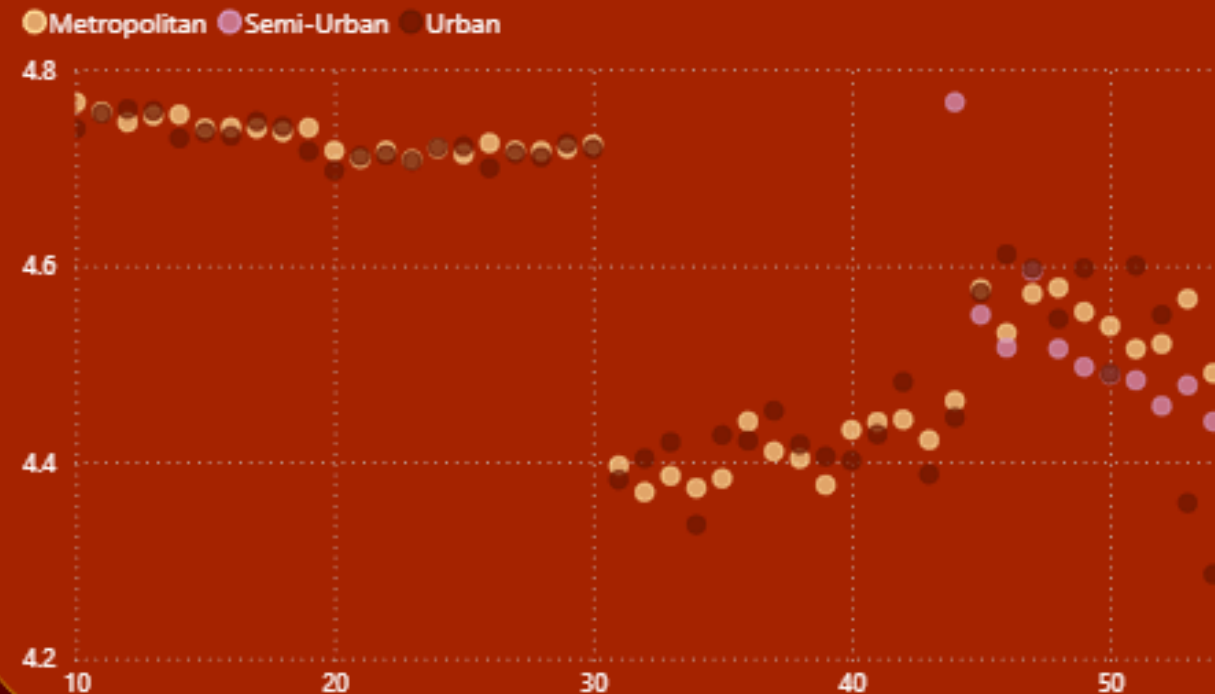
WoW AVG Rating



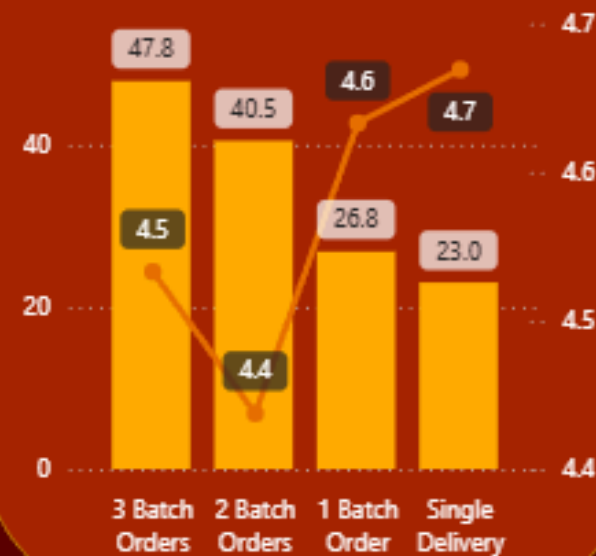
Delivery Performance & Rating Analysis



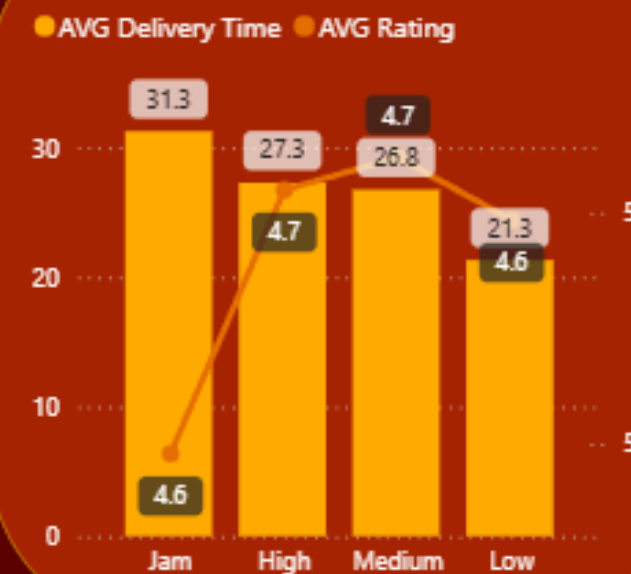
Correlation: Delivery Duration vs. Customer Rating



Multiple Order Impact on Time & Rating



Traffic Impact on Time & Rating



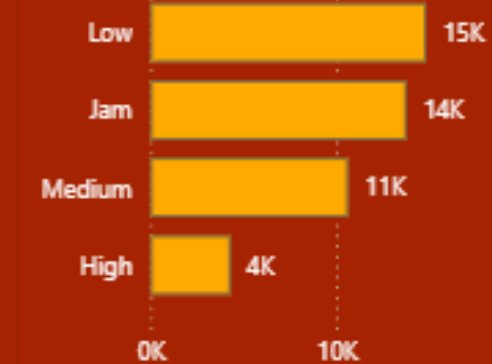
Select Festival:

Select City:

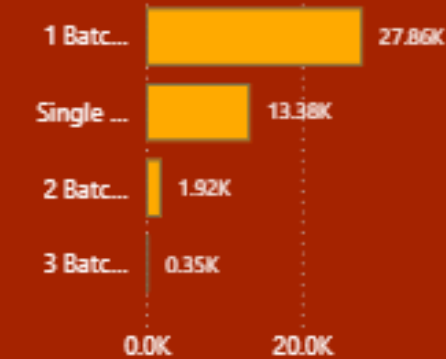
All

All

Road Traffic Density



Multiple Delivery Share

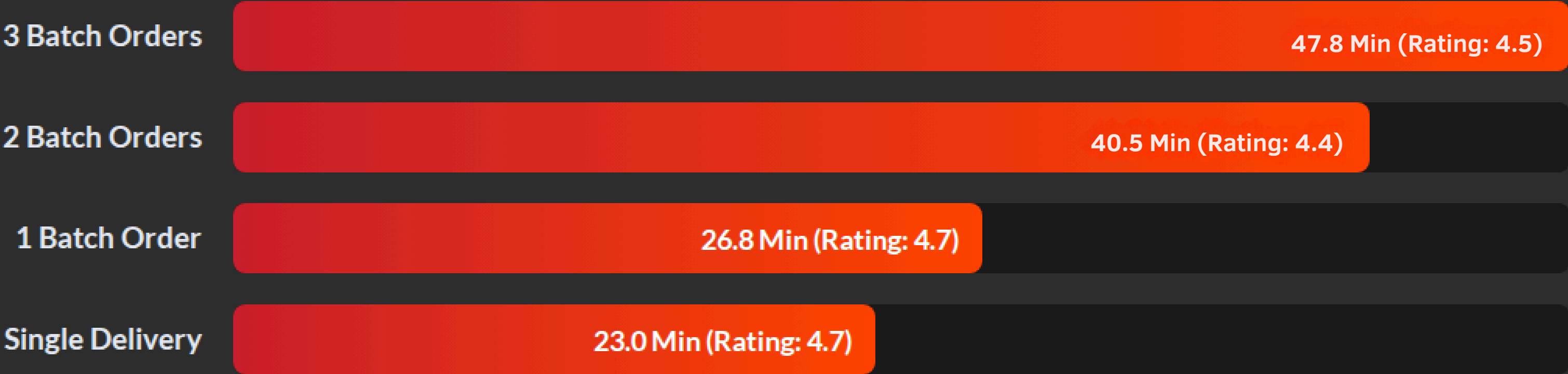


Insight:

- Customer ratings remain high (~4.7-4.8) for deliveries under 30 minutes, but drop sharply to ~4.4-4.5 as duration exceeds 30 minutes.
- System assignment of 3 Batch Orders drastically inflates delivery time to 47.8 mins, driving ratings down to an all-time low of 4.4

DELAY CAUSE #1: MULTIPLE BATCHING

Average Delivery Time by Batch Type



Insight: Systematically assigning drivers multiple concurrent orders severely exacerbates delays. Any delivery extending past 30 minutes destroys the customer rating.

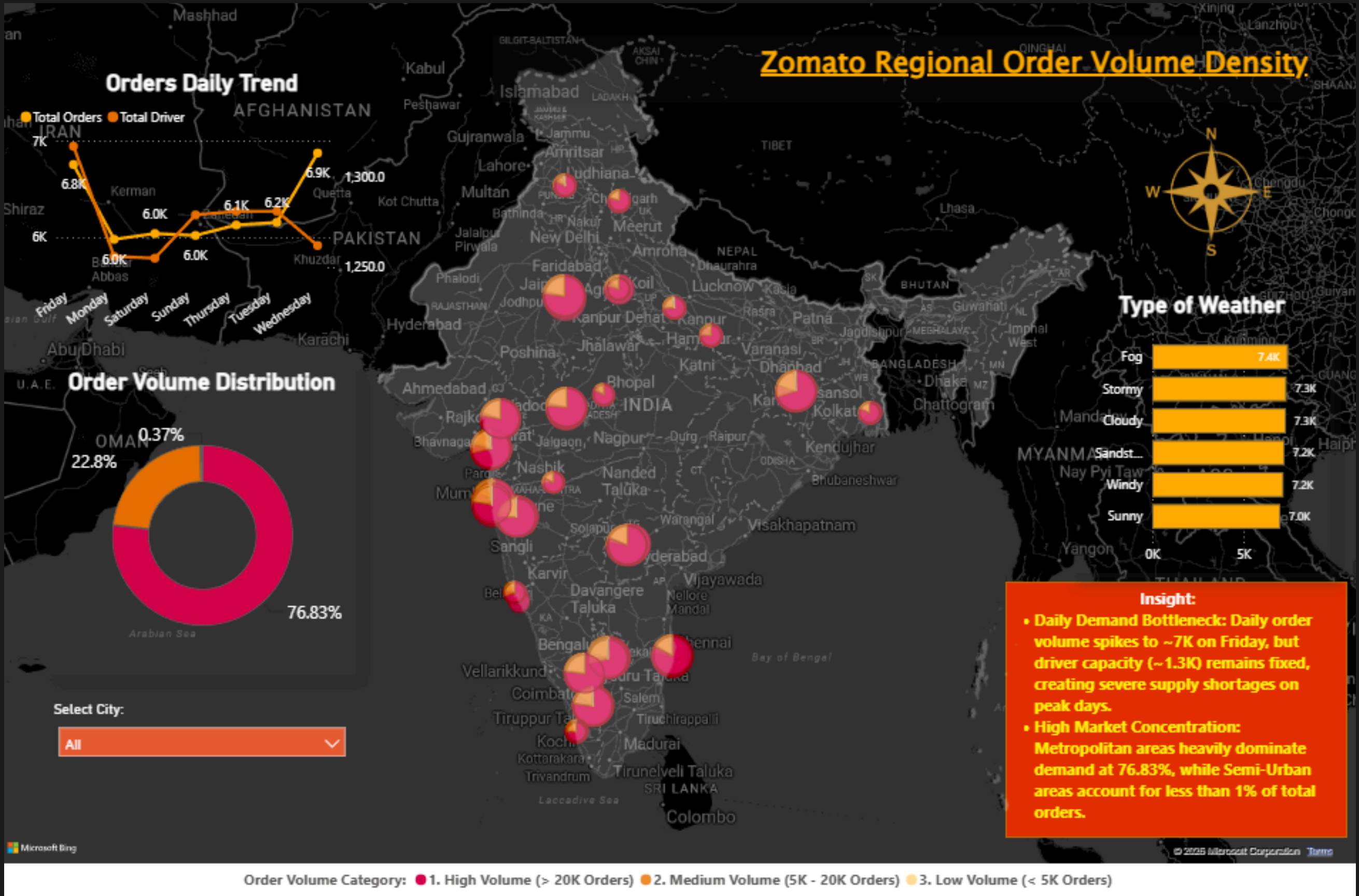
DELAY CAUSE #2: TRAFFIC DENSITY

Average Delivery Time by Traffic Status



Insight: Severe traffic density breaches the critical 30-minute Service Level Agreement (SLA), instantly dropping the average rating.

Preview Dashboard(3/3)



THE GEOGRAPHIC **IMBALANCE**

City Classification	Total Orders	Active Drivers	Avg. Time	Operational Status
Metropolitan	34,000	1,320	27.3 Min	Massive Demand, Overwhelmed Fleet
Urban	10,000	1,281	23.0 Min	Low Demand, Underutilized Fleet
Semi-Urban	< 1,000	154	49.7 Min	Critical Danger Zone

*There is a severe misallocation of resources. Urban areas have almost the same number of drivers as Metro, but handle 3x fewer orders.

STRATEGIC ACTION PLAN: QUICK WINS

Dynamic Batching Restriction

Cap the system at **1 Batch Order** maximum to maintain delivery times safely under 27 minutes.

Temporarily **disable** 2 & 3 Batch features during peak surge hours (17:00-00:00) or whenever live traffic status is flagged as 'Jam'.

SLA Warning System

If predictive logistics anticipate a delay exceeding 30 minutes, proactively issue a 'Free Delivery Voucher' in-app to secure customer retention.



STRATEGIC ACTION PLAN: LONG TERM



Fleet Reallocation

Urban areas are overstaffed (1.2K drivers for only 10K orders). We must create targeted incentives to **transfer Urban drivers into Metropolitan zones**.

Alternatively, launch an aggressive hiring campaign specifically geofenced for Metropolitan areas to bridge the gap.

Predictive Logistics Hubs

To combat the massive volume of 'Long Distance' orders in Metro areas, establish strategic Cloud Kitchens in identified traffic blind spots.

EXECUTIVE SUMMARY

- **The Anomaly:** Order volumes experienced extreme stagnation throughout March, accompanied by unexplained drops in Week 2.
- **The Root Cause:** Data correlation proves that daily order volume is directly driven by Customer Ratings (+0.96). When ratings drop, orders immediately stagnate.
- **The Bottleneck:** Ratings are severely dragged down by delayed deliveries (-0.98), primarily caused by Traffic Jams and Inefficient 3-Batch Orders.
- **The Solution:** Restrict batching during peak hours and reallocate underutilized Urban drivers to overwhelmed Metropolitan areas.



Thank You

[link resource file Power BI & Google Colab....](#)

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